ANALYSIS REPORT

Track 2 Forward Reference Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares five forward `Track 2` reconstructed-curve candidates on the same `97` held-out forward curves:

- `paper_original_best_Fw`, from repository paper-original reference banks;
- `paper_retuned_best_Fw`, from repository paper-retuned reference banks;
- `paper_original_best_Fw_original_onnx_release`, loaded directly from the recovered original `ONNX` release;
- `rcim_original_simplified_onnx_Fw`, using harmonics `0`, `1`, `39`, and `40`;
- `rcim_original_plc_hgbm_onnx_Fw`, using only `HGBM` original `ONNX` models for those sparse harmonics.

  The collages below are regenerated with the same four representative forward curves for every candidate, so visual differences are directly comparable across models. Model labels used in compact tables:

Label	Candidate
Original	`paper_original_best_Fw`
Retuned	`paper_retuned_best_Fw`
Full ONNX	`paper_original_best_Fw_original_onnx_release`
Sparse	`rcim_original_simplified_onnx_Fw`
PLC HGBM	`rcim_original_plc_hgbm_onnx_Fw`

Aggregate Track 2 Metrics

Candidate	Curves	MAE [deg]	RMSE [deg]	Mean Error [%]	P95 Error [%]
`paper_original_best_Fw`	97	0.002768839	0.002951263	6.250311	13.826566
`paper_retuned_best_Fw`	97	0.001839362	0.002040887	4.108527	9.865680
`paper_original_best_Fw_original_onnx_release`	97	0.002803544	0.002986894	6.329387	13.846840
`rcim_original_simplified_onnx_Fw`	97	0.002617089	0.002979175	5.729772	11.357307
`rcim_original_plc_hgbm_onnx_Fw`	97	0.002449482	0.002809453	5.338223	10.932099

Pairwise Predicted-Curve Differences

Pair	Mean MAE [deg]	P95 [deg]	Max [deg]	RMSE [deg]	Corr.
Original vs Retuned	0.001854789	0.004372766	0.006321356	0.002361028	0.999986565
Original vs Full ONNX	0.000202431	0.000491746	0.000636939	0.000247559	0.999992899
Original vs Sparse	0.002338454	0.007941779	0.019313790	0.003122241	0.988188713
Original vs PLC HGBM	0.002249346	0.007310691	0.020324662	0.002959511	0.988187717
Retuned vs Full ONNX	0.001906367	0.004418687	0.006470047	0.002417028	0.999992636
Retuned vs Sparse	0.001998912	0.006889777	0.019046146	0.002751516	0.988208626
Retuned vs PLC HGBM	0.001337372	0.005218275	0.019153349	0.002094972	0.988208183
Full ONNX vs Sparse	0.002368388	0.008159896	0.019655131	0.003161418	0.988187282
Full ONNX vs PLC HGBM	0.002285644	0.007483631	0.020666003	0.003000129	0.988186285
Sparse vs PLC HGBM	0.001391881	0.004152736	0.005587220	0.001830819	0.999999000

Collage Curve Metrics

All metric columns in this section are `MAE [deg]`.

Curve	Operating Point	Original	Retuned	Full ONNX	Sparse	PLC HGBM
`Curve 1`	100 rpm / 100 Nm / 25 C	0.003931161	0.003849563	0.004471498	0.004241496	0.004190061
`Curve 2`	1600 rpm / 1500 Nm / 25 C	0.001674670	0.001472680	0.001934975	0.003028629	0.002887111
`Curve 3`	700 rpm / 1400 Nm / 30 C	0.001514636	0.001533115	0.001358691	0.004597120	0.002192193
`Curve 4`	800 rpm / 1800 Nm / 35 C	0.001046320	0.001916711	0.001279273	0.003744366	0.002411656

Collage Curve Anchor Deltas

All delta columns in this section are predicted-curve `MAE [deg]`.

Curve	Operating Point	Full ONNX	Sparse	PLC	PLC-Full
`Curve 1`	100 rpm / 100 Nm / 25 C	0.000540337	0.000526955	0.000508614	0.000505684
`Curve 2`	1600 rpm / 1500 Nm / 25 C	0.000444175	0.002496205	0.002530544	0.002614484
`Curve 3`	700 rpm / 1400 Nm / 30 C	0.000162021	0.003089332	0.001249842	0.001317514
`Curve 4`	800 rpm / 1800 Nm / 35 C	0.000293090	0.002855756	0.001800590	0.001673647

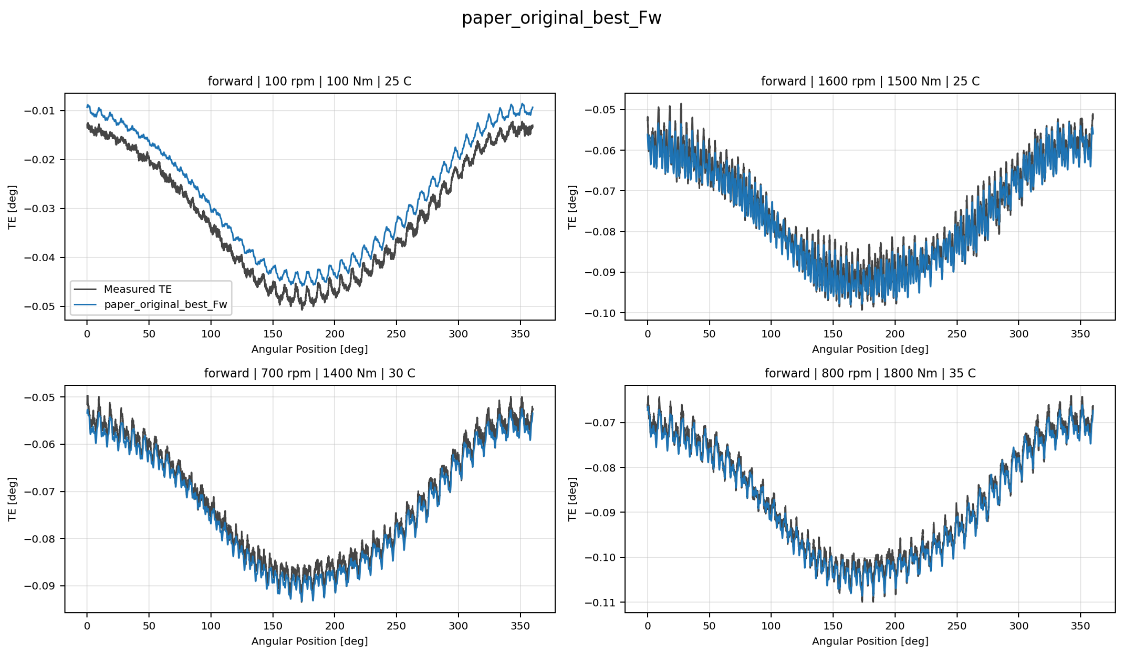
Technical Interpretation

The full recovered original `ONNX` release remains almost superposed with `paper_original_best_Fw`: the pair has the smallest mean curve-difference `MAE`, a near-unit mean correlation, and only small aggregate metric changes. This supports the same conclusion as the earlier diagnostic: the repository paper-original bank and the recovered original `ONNX` release are effectively the same Track 2 reconstructed surface, with minor differences attributable to archive/export/loading path details.

`paper_retuned_best_Fw` is visually shape-aligned with the paper-original surface, but it is numerically distinct and improves the measured-curve metrics. The sparse original `ONNX` variants are also shape-aligned, but they move away from the 19-target original surface because they retain only harmonics `0`, `1`, `39`, and `40`. In this run, the PLC-oriented all-`HGBM` sparse variant is the stronger sparse candidate.

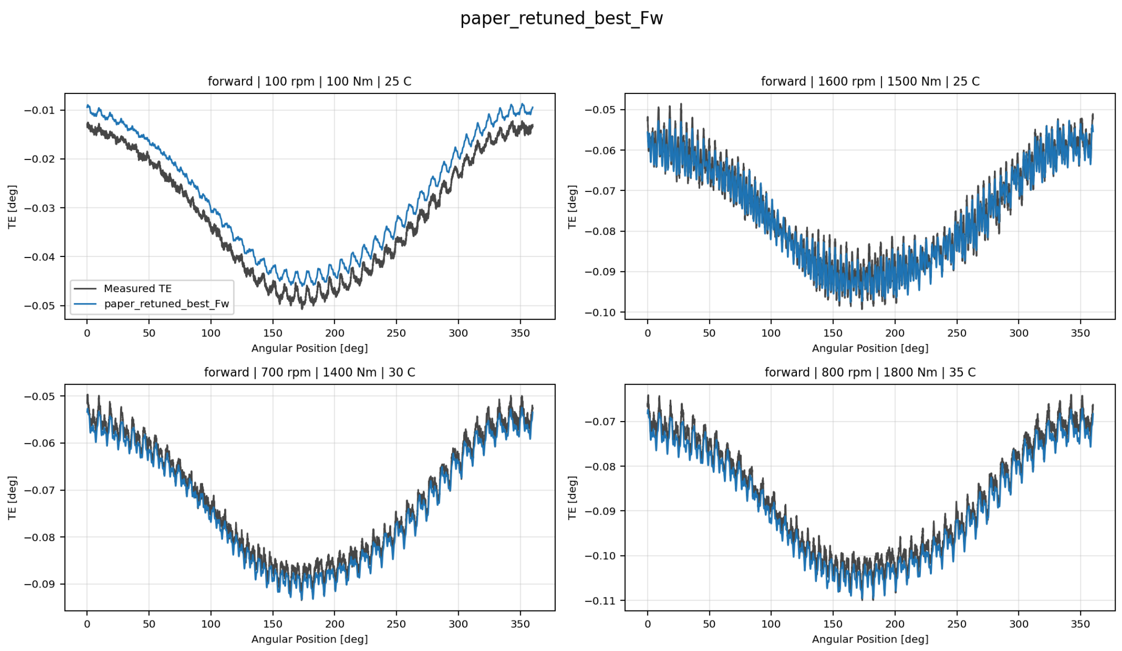
Candidate Collages

paper_original_best_Fw



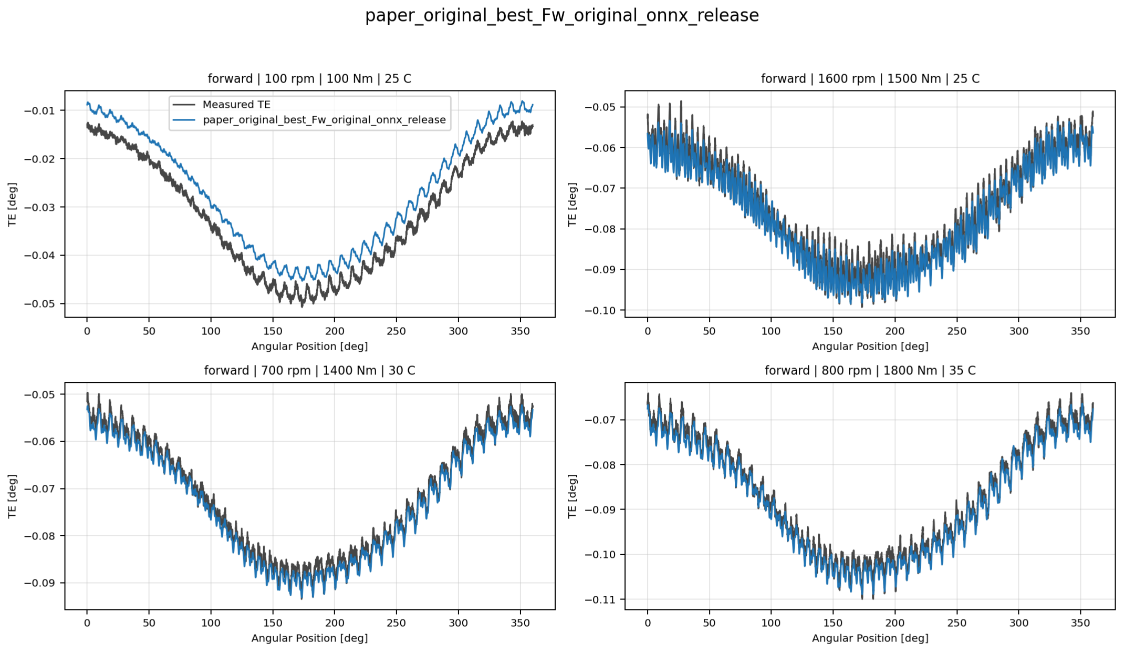
paper_original_best_Fw Track 2 collage

paper_retuned_best_Fw



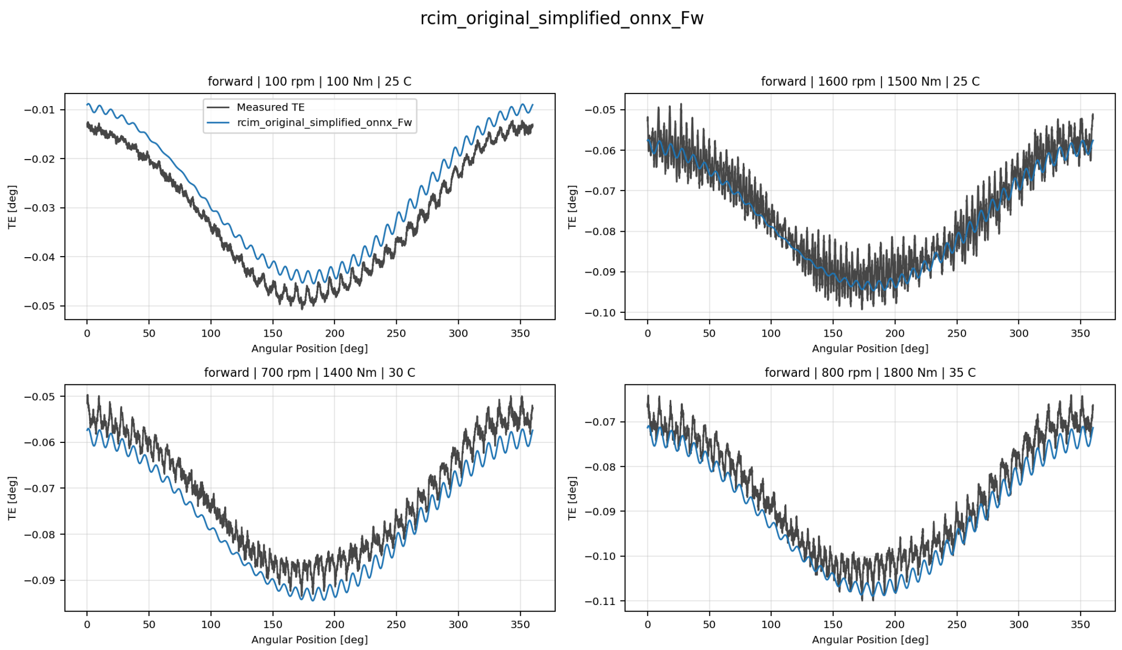
paper_retuned_best_Fw Track 2 collage

paper_original_best_Fw_original_onnx_release



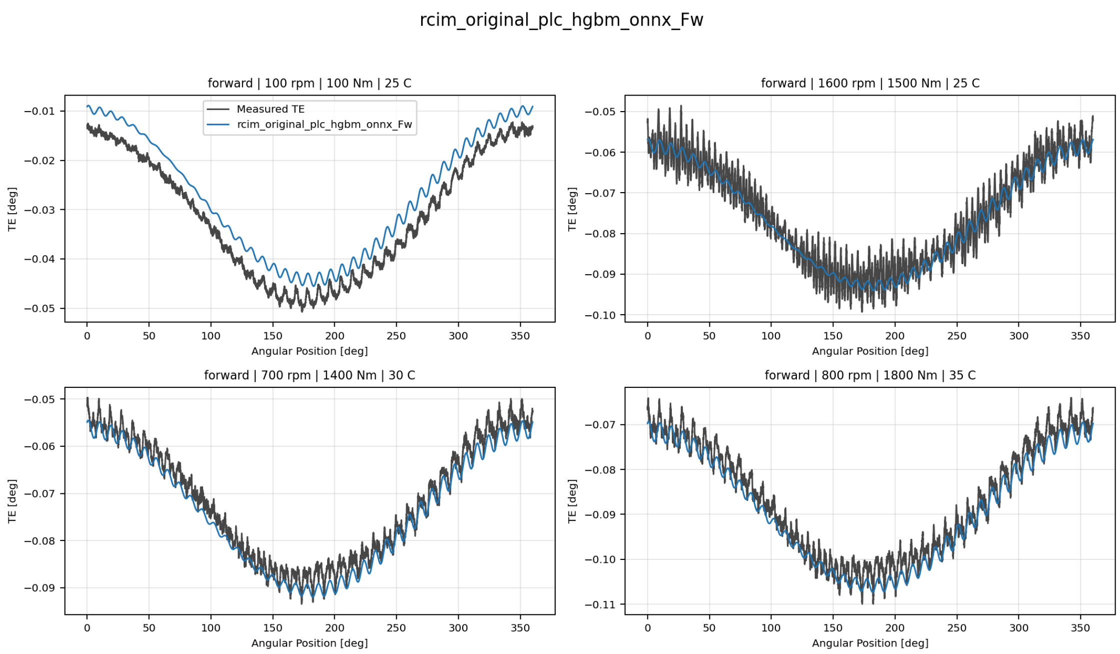
paper_original_best_Fw_original_onnx_release Track 2 collage

rcim_original_simplified_onnx_Fw



rcim_original_simplified_onnx_Fw Track 2 collage

rcim_original_plc_hgbm_onnx_Fw



rcim_original_plc_hgbm_onnx_Fw Track 2 collage

Output Artifacts

- output directory: `output\validation_checks\track2_forward_reference_curve_comparison\2026-06-08-18-00-50__track2_forward_reference_curve_comparison` ;
- summary YAML: `output\validation_checks\track2_forward_reference_curve_comparison\2026-06-08-18-00-50__track2_forward_reference_curve_comparison\track2_forward_reference_curve_comparison_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_forward_reference_curve_comparison\2026-06-08-18-00-50__track2_forward_reference_curve_comparison\track2_forward_reference_curve_comparison_metrics.csv` ;
- pairwise CSV: `output\validation_checks\track2_forward_reference_curve_comparison\2026-06-08-18-00-50__track2_forward_reference_curve_comparison\track2_forward_reference_curve_pairwise_differences.csv` ;
- report Markdown: `doc\reports\analysis\track2\forward_reference_curve_comparison\[2026-06-08]\track2_forward_reference_curve_comparison_report.md` .